C Hemish Reddy

es24btech11012@iith.ac.in | hemishreddychittibala@gmail.com | +91 6366972226 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.57
Sri Chaitanya Techno School Marathalli Bangalore	Class XII (CBSE)	2024	94.00%
Air Force School Jalahalli Bangalore	Class X (CBSE)	2022	89.00%

SKILLS

Languages: C programming, C++, Python, SQL.

Libraries: NumPy, Pandas, Matplotlib.

Technologies: Git/GitHub, MySQL.

Core Concept: DSA, OOPs, Operating Systems.

Soft Skills: Communication, Critical Thinking, Teamwork,Adaptability,Time management

PROJECTS

Mini Unix Shell | Github

- Developed an **interactive Unix-like command-line interpreter** in C featuring a robust execution loop, custom environment variable management (export/unset), and state-aware directory navigation (cd -).
- Designed a **quote-aware lexical analyzer and parser** capable of processing whitespace-separated tokens, single/double quotes, and unspaced shell operators.
- Managed **multi-process lifecycles** using POSIX system calls including fork(), execvp(), and waitpid(), enabling process isolation, synchronous foreground execution, and exit status tracking.
- Implemented **multi-stage pipelines and file descriptor redirection** using pipe() and dup2() to support input (<), output (>), append (>>), and arbitrary-depth command chaining.

Technologies / Tools Used : C, Make, POSIX, Linux

Search Engine in C++ | GitHub

- Architected a single-file prototype into a **modular C++17 search engine** with independent subsystems for web crawling, storage, indexing, and ranking.
- Integrated **libcurl** for live web crawling from seed configuration files and built a custom **HTML parser** using a Document Object Model (DOM) tree for structured text extraction.
- Designed an **Inverted Index** using tokenization, stop-word removal, and custom word stemming to optimize term-to-document posting list lookups.
- Implemented a **TF-IDF-based ranking pipeline** for relevance scoring, achieving sub-second query lookups across indexed document repositories.

Technologies / Tools Used : C++, libcurl, CMake, STL

Distributed In-Memory Cache Engine | Github

- Built a production-style **in-memory key-value cache engine** with a modular architecture separating storage, eviction, expiration, persistence, and protocol layers.
- Implemented a **thread-safe KV store** using std::unordered_map and std::shared_mutex to support concurrent read/write operations.
- Designed **TTL-based expiration** with a background cleanup worker and implemented **O(1) LRU eviction** using a hash map and doubly linked list.
- Developed **snapshot-based persistence** for cache recovery and added **unit testing with GoogleTest** to validate core functionality.

- Architected the engine as a foundation for a future **distributed cache**, supporting consistent hashing, replication, and multi-node deployment.

Technologies/Tools used: C++20 , STL , Multithreading , Shared Mutex , CMake , GoogleTest

Interprocess Producer-Consumer System using Shared Memory and Semaphores

- Built a multi-process **Producer-Consumer System** in **C** using **POSIX Shared Memory** and **POSIX Semaphores**.

- Implemented **process synchronization** mechanisms to prevent race conditions between concurrent processes.

- Designed a **Circular Buffer** for efficient shared resource management and interprocess communication.

- Coordinated safe execution across processes created using **fork()**, ensuring correct producer-consumer behavior.

RELEVANT COURSES

Core Course: Introdution to programming, Algorthims, Data Structure and their Applications, Operating Systems, Theory of Computation , Discrete Mathematics for Computer Science.

Math Course: Calculus I, Calculus II, Probability and Random Variables, Intro to Statistics, Differential Equations, Linear Algebra, Complex Variables.

POSITIONS OF RESPONSIBILITY

Coordinator | Media Council Logistics department, IIT Hyderabad

- Coordinated logistics operations for institute merchandise distribution and campus events
- Managed workflow execution across multiple student activities and club operations
- Collaborated with teams to ensure smooth event coordination and timely delivery.

EXTRACURRICULAR

- **NSO Volleyball player**
- Won silver in **Volleyball** in MILAN (Inter Hostel Competion)